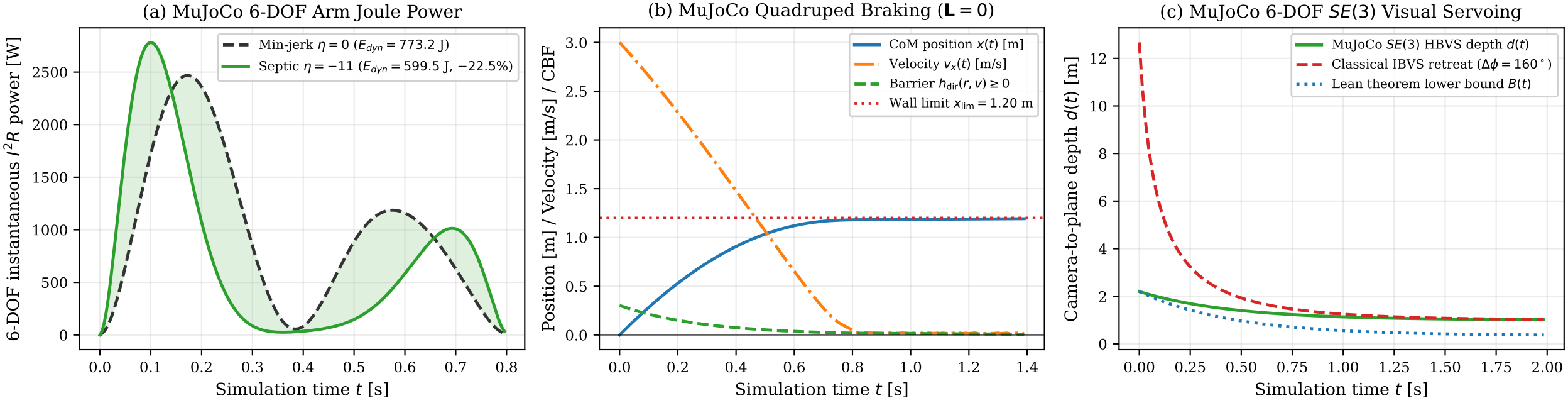
(a) MuJoCo 6-DOF Arm Joule Power
6-DOF instantaneous I^2R power [W]
Simulation time t [s]
Min-jerk $\eta = 0$ ($E_{dyn} = 773.2$ J)
Septic $\eta = -11$ ($E_{dyn} = 599.5$ J, -22.5%)
(b) MuJoCo Quadruped Braking ($\dot{\mathbf{L}} = 0$)
Position [m] / Velocity [m/s] / CBF
Simulation time t [s]
CoM position $x(t)$ [m]
Velocity $v_x(t)$ [m/s]
Barrier $h_{\text{dir}}(r, v) \geq 0$
Wall limit $x_{\text{lim}} = 1.20$ m
(c) MuJoCo 6-DOF $SE(3)$ Visual Servoing
Camera-to-plane depth $d(t)$ [m]
Simulation time t [s]
MuJoCo $SE(3)$ HBVS depth $d(t)$
Classical IBVS retreat ($\Delta\phi = 160^\circ$)
Lean theorem lower bound $B(t)$